

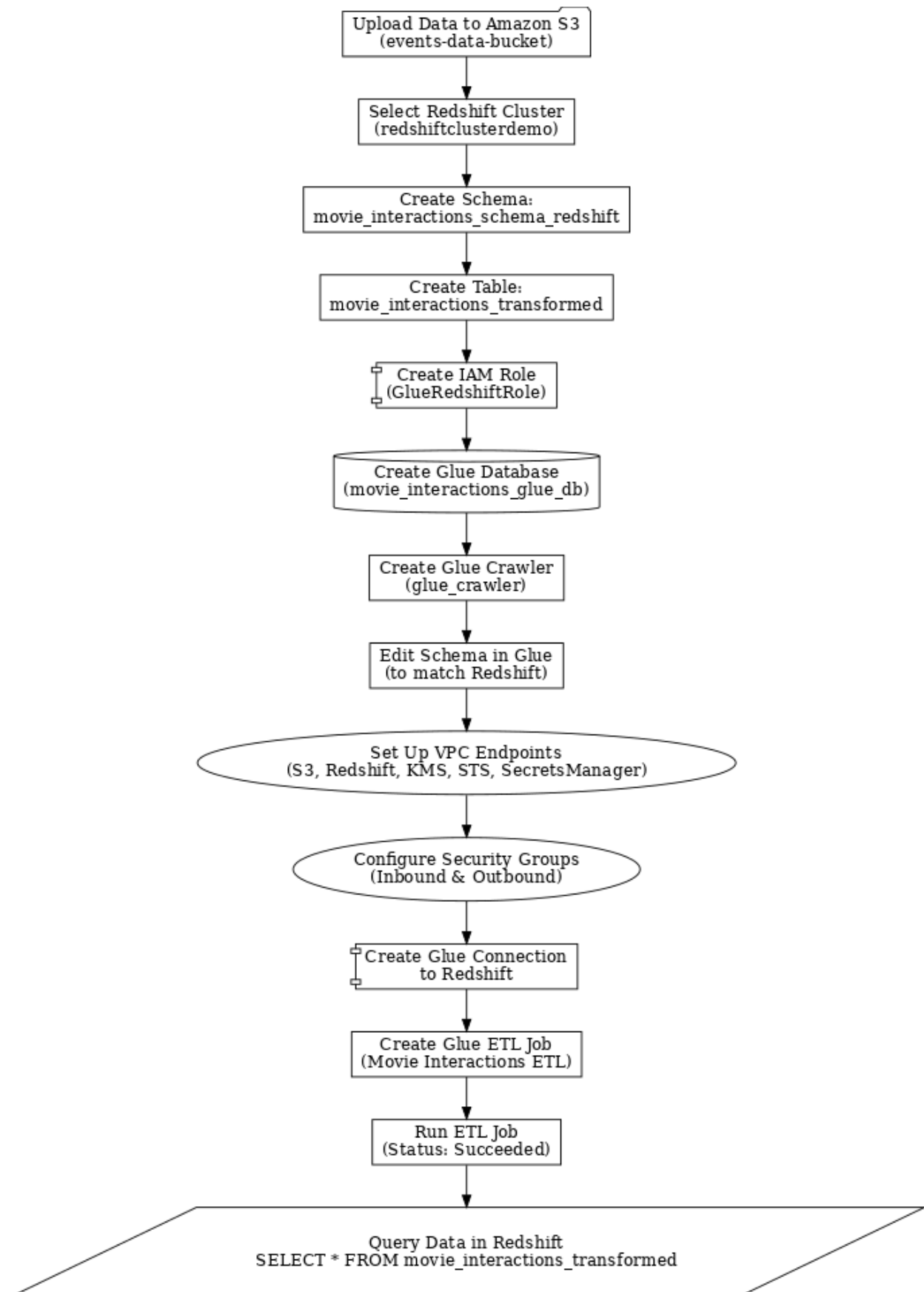


table).

So the flow is:

” Ask ChatGPT

Copy code

```
S3 (CSV) --[Crawler→Data Catalog]--> Glue ETL (transform) --[Glue connection]--> Redshift table
```

You then query the cleaned table from Redshift.

REDSHIFT:

Amazon Redshift > Clusters > Edit admin credentials

Edit admin credentials

Change admin user password

Admin password
Select an option to manage your admin password.

☐ Manage admin credentials in AWS Secrets Manager [Info](#)

☒ **Generate a password**
Amazon Redshift generates an admin password.

New admin user password

XXXXXXXXXX

☐ Show password

☐ Manually add the admin password
Manually enter the admin password.

Cancel **Save changes**

Save the password and we have to use in glue

CRAWER:

WS Glue

glue_crawler

September 28, 2025 at 04:42:11

Run crawler

Crawler properties

Name	glue_crawler	IAM role	Glueredshiftrole	Database	movie_interactions_glue_db	State	READY
Description	-	Security configuration	-	Lake Formation configuration	-	Table prefix	raw_
Maximum table threshold	-						

Advanced settings

Crawler runs | Schedule | Data sources | Classifiers | Tags

Data sources (1)

The list of data sources to be scanned by the crawler.

Type	Data source	Parameters
------	-------------	------------

The screenshot shows the AWS Glue console interface. At the top is a navigation bar with a search box containing "Search", a keyboard shortcut "[Alt+S]", and icons for notifications, help, settings, and a dropdown menu set to "United States (N. Virginia)". Below the navigation bar is a breadcrumb trail: "Home > Crawlers > sampe". On the left side, there is a sidebar with a search icon and a list of items: "e", "rted", "TL", "yks", "monitoring", "ig tables", "ctions", "(orchestration)", "tegrations New", "log", "ema registries", "s", and "s". The main content area displays the "Crawler properties" for a crawler named "sampe". It includes fields for Name, Description, Maximum table threshold, IAM role (Glueredshiftrole), Security configuration, Database (movie_interactions_glue_db), Lake Formation configuration, State (READY), and Table prefix (raws_). Below these properties is an "Advanced settings" section. At the bottom, there are tabs for "Crawler runs", "Schedule", "Data sources", "Classifiers", and "Tags". The "Crawler runs" tab is active, showing "Crawler runs (1/1)" and a description: "The list of crawler runs for this crawler." There are also buttons for "Stop run" and "View CloudWatch logs".

Crawler properties

Name sampe	IAM role Glueredshiftrole	Database movie_interactions_glue_db	State READY
Description -	Security configuration -	Lake Formation configuration -	Table prefix raws_
Maximum table threshold -			

► Advanced settings

Crawler runs | Schedule | Data sources | Classifiers | Tags

Crawler runs (1/1)

The list of crawler runs for this crawler.

[Refresh] [Stop run] [View CloudWatch logs]

🔔 **Announcing new optimization features for Apache Iceberg tables**
Optimize storage for Apache Iceberg tables with automatic snapshot retention and orphan file deletion. [Learn more](#) 📄

movie_interactions_glue_db

Last updated (UTC)
September 28, 2025 at 04:48:30  [Edit](#)

Database properties

Name	Description	Location	Created on (UTC)
movie_interactions_glue_db	-	-	September 28, 2025 at 03:5

Tables (2)

Last updated (UTC)
September 28, 2025 at 04:51:47  [Delete](#) [Add tables using crawler](#)

View and manage all available tables.

☐	Name	Database	Location	Classification	Deprecated	View data	Data quality	Color
☐	raw_events_data_bi	movie_interactions	s3://events-data-bi	CSV	-	Table data	View data quality	View

Edit schema to match redshift schema in table in glue db

us-east-1.console.aws.amazon.com/glue/home?region=us-east-1#/v2/data-catalog/tables/edit-schema-json/

Search [Alt+S]

Tables > raws_events_data_bucket9c1c > Edit schema as JSON

```
7    "Name": "user_id",
8    "Type": "int",
9  },
10 {
11   "Name": "movie_id",
12   "Type": "int",
13 },
14 {
15   "Name": "interaction_type",
16   "Type": "string",
17 },
18 {
19   "Name": "event_timestamp",
20   "Type": "string",
21 },
22 {
23   "Name": "device_type",
24   "Type": "string",
25 },
26 {
27   "Name": "session_id",
28   "Type": "string",
29 },
30 {
31   "Name": "duration_seconds",
32   "Type": "bigint",
33 },
34 ]
```

INT

Create vpc endpoints and edit security groups:

Endpoints (6) Info

Find endpoints by attribute or tag

Name	VPC endpoint ID	Endpoint type	Status	Service
s3-gateway-endpoint	vpce-02c66b702bbbe779a	Gateway	Available	com.amazonaws.s3
redshift-interface-endpoint	vpce-0c74dec2f2e04b9c9	Interface	Available	com.amazonaws.redshift
sts-interface-endpoint	vpce-01cf5e74fc2e7cabe	Interface	Available	com.amazonaws.sts
secretsmanager-interface-endpoint	vpce-071d250cdb0936396	Interface	Available	com.amazonaws.secretsmanager

Select an endpoint

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sgr-0498c251c7f6abcf	All TCP	TCP	0 - 65535	Custom	

Add rule

Cancel Preview changes

Edit outbound rules Info

Outbound rules control the outgoing traffic that's allowed to leave the instance.

Outbound rules Info

Security group rule ID	Type	Protocol	Port range	Destination	Description - optional
sgr-099e67e1ee940930f	All TCP	TCP	0 - 65535	Custom	
sgr-0cc32454a6d6a4309	HTTPS	TCP	443	Custom	

Add rule

Cancel Preview changes Save

Glue connection to connect to redshift:

10. Navigate back to **Endpoints**, and wait until all endpoints have a **Status of Available**. Click **Refresh** periodically until they do.

Create a Glue connection

1. Navigate to **AWS Glue**, toward the right side, a bit further down, click on **Go to connections**, then click on **Create connection**.
2. For the Data source, select **Amazon Redshift**, and in the bottom right click **Next**.
3. At the **Configuration connection** page, fill in the following values:
 - **Database instances:** Select **redshiftclusterdemo**.
 - **Password:** Paste in the password that you stored earlier.
 - **IAM service role:** Select the **GlueRedshiftRole**.
4. Click **Next**, then click **Next** again.
5. Click **Create connection**.

Use password here

The screenshot shows the 'Configure connection' page in the AWS Glue console. On the left, a sidebar lists navigation options: 'Connectors', 'Data sources', 'Jobs', 'Crawlers', 'Databases', 'Tables', 'Partitions', 'Queries', 'Dashboards', 'Alerts', 'Connections', 'Endpoints', 'Vendors', 'Catalog', 'Data stores', 'Data sources', 'Jobs', 'Crawlers', 'Databases', 'Tables', 'Partitions', 'Queries', 'Dashboards', 'Alerts', 'Connections', 'Endpoints', 'Vendors', 'Catalog', 'Data stores'. The main content area is titled 'Configure connection' and contains the following fields:

- Database instances:** A dropdown menu showing 'redshiftclusterdemo-ccnc3dm3d3sy' with a refresh icon.
- Database name:** A text input field containing 'interactions_db'.
- Credential type:** Two radio buttons: 'Username and password' (selected) and 'AWS Secrets Manager'.
- Username:** A text input field containing 'admin'.
- Password:** A password input field with masked characters.
- Enforce SSL - optional:** An unchecked checkbox.
- IAM service role:** A dropdown menu showing 'GlueRedshiftRole' with a refresh icon.

At the bottom right, there are 'Cancel' and 'Previous' buttons.

Search[Alt+S]

United States (N. Virginia)

mviet

Last modified on 9/28/2025, 12:25:46 AMActions

VisualScriptJob detailsRunsData qualitySchedulesVersion Control

+Data source - Data CatalogAWS Glue Data Catalog

Data previewOutput schema

Data preview (16)InfoREADY

Filter sample dataset

interaction_id	user_id	movie_id	interaction_type	event_timestamp	device_type
1	101	1001	click	2025-02-02 10:00:00	mobile
1	101	1001	click	2025-02-02 10:00:00	mobile
2	102	1002	view	2025-02-03 11:00:00	tablet

End sessionPreviewing 8 of 8 fields

Data source properties - Data Catalog

NameAWS Glue Data Catalog

DatabaseChoose a database.
movie_interactions_glue_db

Tableraw_events_data_bucket9c1c

Partition predicate - optionalEnter a boolean expression supported by Spark SQL, using only partition columns.
Partition predicate syntax for Spark SQL is year == year[date_sub(current_date, 7)] AND month(date_sub(current_date, 7)) AND day == day[date_sub(current_date, 7)].

data

et2

VisualScriptJob detailsRunsData qualitySchedulesVersion Control

+Data source - Data CatalogAWS Glue Data Catalog

Data previewOutput schema

Data preview (16)InfoREADY

Filter sample dataset

interaction_id	user_id	movie_id	interaction_type	event_timestamp	device_type
1	101	1001	click	2025-02-02 10:00:00	mobile
1	101	1001	click	2025-02-02 10:00:00	mobile
2	102	1002	view	2025-02-03 11:00:00	tablet

End sessionPreviewing 8 of 8 fields

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Partition predicate syntax for Spark SQL is year == year[date_sub(current_date, 7)] AND month(date_sub(current_date, 7)) AND day == day[date_sub(current_date, 7)].

You are changing schema and removing duplicates

The screenshot shows the AWS Glue Studio console interface. The top navigation bar includes tabs for Script, Job details, Runs, Data quality, Schedules, and Version Control. The main workspace displays a workflow with two transforms: 'Transform - Change Schema' and 'Transform - Drop Duplicates', both marked with green checkmarks. A 'Data target - Amazon Redshift' node is at the bottom. On the left, a sidebar lists various data sources and targets, including Glue Data Catalog, Amazon S3, Amazon Redshift, and various SQL databases. On the right, the 'Data target properties - Amazon Redshift' panel is open, showing fields for Name, Node parents, Redshift access type (set to 'Direct data connection - recommended'), Redshift connection, and Connection. A red warning message at the top right states 'Job has not been saved'.

And load data to redshift from glue

The screenshot shows the AWS Glue Studio console interface. The top navigation bar includes tabs for Visual, Script, Job details, Runs, Data quality, Schedules, and Version Control. The main workspace displays a workflow with a single node: 'Data target - Amazon Redshift'. On the right, the 'Redshift connection' panel is open, showing fields for Redshift connection, Connection name, Schema, and Table. The 'Handling of data and target table' section is also visible, with options for APPEND (insert) to target table, MERGE data into target table, TRUNCATE target table, and DROP and recreate target table. A red warning message at the top right states 'Job has not been saved'. A blue error message at the bottom left states 'Target node not supported. You have selected a data target node which is not supported for data preview. Please select another type of node instead.'

Check run:

Successfully started job et2. Navigate to Run details for more details.

et2

Last modified on 9/28/2025, 1:01:29 AM

Visual Script Job details Runs Data quality Schedules Version Control

Data target - Amazon Redshift

Data target properties - Amazon Redshift

Name: Amazon Redshift

Node parents: Choose which nodes will provide inputs for this one. Choose one or more parent node

Drop Duplicates

Redshift access type: Direct data connection - recommended

Redshift connection: Choose the AWS Glue connection for Amazon Redshift, or create a new connection

Connection: Database

Target node not supported

You have selected a data target node which is not supported for data preview. Please select another type of node instead.

In redshift:

us-east-1.console.aws.amazon.com/redshiftv2/home?region=us-east-1#query-editor:

Run Save Schedule Clear

Send feedback

Query results Table details

Query 4185

Completed, started on September 28, 2025 at 01:02:29

ELAPSED TIME: 00 m 02 s

Rows returned (10)

Export

Search rows

interaction_id	user_id	movie_id	type	event_time	device_type	session_id	duration_seconds
1	101	1001	click		mobile	session_001	120
2	102	1002	view		tablet	session_002	300
3	101	1001	click		mobile	session_003	180
4	103	1003	view		desktop	session_004	150
5	104	1004	click		mobile	session_005	220
6	105	1005	view		tablet	session_006	250
7	106	1006	click		desktop	session_007	200
8	107	1007	view		mobile	session_008	310
9	108	1008	click		tablet	session_009	130